

FYP Template Pack

A Student's Guide to the FYP-I and FYP-II LaTeX Templates

This guide explains the four ready-to-use LaTeX templates provided for the Final Year Project (FYP-I and FYP-II), covering both research-based and application-based projects, and walks you through choosing the right one, filling it in, citing sources, and compiling it into a PDF.

1. What's in the Template Pack

There are four separate templates. Each is a self-contained folder — download the one your project needs and work inside it.

Template	Use for	Format
FYP-I Research-Based Proposal	FYP-I proposal, if your project investigates a research question	Single .tex document (~12 pages)
FYP-I Application-Based Proposal	FYP-I proposal, if your project builds a software system/product	Single .tex document (~13 pages)
FYP-II Research-Based Report	FYP-II final report, research track	Multi-chapter book-style document (~29 pages)
FYP-II Application-Based Report	FYP-II final report, application/development track	Multi-chapter book-style document (~32 pages)

All four share the same University of Lahore, Sargodha Campus branding, colour palette, and fonts, so your FYP-I proposal and FYP-II report will look like a consistent, professional pair of documents.

2. Which Template Should I Use?

Ask two questions: (1) which phase am I in, and (2) is my project centred on a research question or on building a working system?

2.1 Research-based vs. application-based

If your project...	...is
Investigates a hypothesis or research question, compares algorithms/methods, and evaluates results against a baseline with metrics	Research-based
Designs and builds a working web, mobile, or desktop application for real or realistic users, validated mainly through testing rather than experiments	Application-based

If you're unsure, ask your supervisor early — this decision determines which template (and which checkbox on the Project Registration Form) you use for both FYP-I and FYP-II. Most groups stay on the same track for both semesters.

2.2 Phase

- Registering / proposing your project this semester → use an FYP-I template.
- Writing up the completed project next semester → use the matching FYP-II template.

3. Getting Started

3.1 Option A — Overleaf (recommended, no installation)

1. Go to overleaf.com and create a free account (or log in).
2. Click New Project → Upload Project, and upload the .zip file for your chosen template.
3. Overleaf will unzip it automatically. Open the main .tex file (the one with the long descriptive name, e.g. UOL_FYP1_Research_Proposal_Template.tex) and click Recompile.
4. If your template uses references.bib (both FYP-II templates, and both FYP-I templates), you may need to set the compiler: Menu → Settings → check that a bibliography step runs automatically — Overleaf usually detects this on its own the first time you compile.

3.2 Option B — Local install (TeX Live / MiKTeX)

5. Install a TeX distribution: TeX Live (Windows/Linux/Mac) or MiKTeX (Windows).
6. Unzip the template folder anywhere on your computer.
7. Open a terminal in that folder and run the compile sequence for your template (see Section 7).
8. Open the resulting .pdf to check your work — recompile every time you make changes.

3.3 What's inside each folder

- One main .tex file — this is the only file you open to compile.
- A .cls file (uolfyp.cls or uolfyp2.cls) — the design/formatting engine. You normally never need to edit this.
- Several content .tex files (one per section or chapter) — this is where you type your actual project content.
- references.bib — your citation database (Section 5).
- A figures/ folder — put your own diagrams, screenshots, and charts here.
- README.txt — a quick-reference version of this guide, specific to that template.

4. How to Fill In the Template

4.1 Placeholders

Every spot you need to edit is marked in green bold text inside square brackets, for example:

[Project Title Goes Here]

Search each file for the word placeholder (or just look for green bracketed text) and replace it with your real content. Do not leave any placeholder text in your final submission — examiners will notice.

4.2 Instruction boxes

Grey shaded boxes in italics explain what belongs in that section and roughly how long it should be. These are guidance only — delete each box once you've written the section it describes. They will not appear if you simply leave them in, but a tidy final document shouldn't include them.

4.3 Checkboxes

Checkboxes in the Project Registration Form use plain LaTeX symbols: an empty box is `\Box` and a filled/ticked box is `\blacksquare`. The correct box (Research or Development) is already pre-filled for you depending on which template you downloaded — you only need to change it if your project type genuinely changes.

4.4 Tables and figures

- Tables (comparison tables, requirement tables, timeline tables, test-case tables) are already structured — just replace the placeholder cell text.
- Figures start as a grey dashed placeholder box. Replace the whole `\fbox{...}` block with `\includegraphics[width=0.8\textwidth]{figures/your-image.png}` once you have a real diagram or screenshot, and put the image file itself inside the `figures/` folder.
- Every figure and table has a `\label{...}` — refer to it in your text with `\ref{...}` (e.g. "as shown in Figure~\ref{fig:architecture}") rather than writing "the figure below", since LaTeX figures can shift position when you compile.

5. Working with Citations (references.bib)

All four templates keep citations in a separate file called `references.bib`, instead of typing a reference list by hand. This is the same approach used in real research papers and makes formatting/reordering references automatic.

5.1 Adding a source

Open `references.bib` and add one entry per source, in BibTeX format. For example, for a journal article:

```
@article{smith2024fl,  
  author = {Smith, J. and Doe, A.},  
  title   = {A survey of federated learning},  
  journal = {IEEE Access},  
  volume  = {12},  
  pages   = {100--120},  
  year    = {2024}  
}
```

The word right after the opening brace (`smith2024fl` above) is the citation key — you choose it, and it must be unique in the file.

5.2 Citing it in the text

In any chapter/section file, cite the source using its key:

- `\citep{smith2024fl}` → renders as a numbered, parenthetical citation, e.g. [3]
- `\citel{smith2024fl}` → renders with the number inline in a sentence, e.g. "Smith et al. [3] showed that..."

You do not need to know the reference number yourself — LaTeX assigns and orders numbers automatically based on where each source is first cited, and builds the final reference list for you.

5.3 Where the reference list appears

The FYP-II templates generate the reference list automatically at the end of the last chapter (before the appendices), via `\bibliography{references}`. The FYP-I templates generate it at the end of the References section. You never need to type or number references by hand — just make sure every entry in `references.bib` is cited at least once (unused entries won't be printed by default).

5.4 A note for the FYP-I Research-Based Proposal template

This particular template still uses an inline `\begin{thebibliography}...\end{thebibliography}` block rather than a separate `.bib` file, since it was built first. If you would like it converted to use `references.bib` for consistency with the other three, ask — it's a quick change.

6. Compiling — Exact Commands

If you are using Overleaf, you can usually ignore this section — just click Recompile and Overleaf handles the bibliography step for you. If compiling locally, run the following from inside the template folder, replacing `FILE` with the main `.tex` file's name (without the `.tex` extension):

6.1 Templates WITH a `references.bib` (all four templates)

```
pdflatex FILE.tex
bibtex FILE
pdflatex FILE.tex
pdflatex FILE.tex
```

Four steps, in this exact order. The two final `pdflatex` passes are both needed: the first resolves citation numbers, the second resolves any cross-references (Table/Figure numbers, table of contents) that shifted as a result.

6.2 If you haven't cited anything yet

Running `bibtex` before you've added any `\citep{}` calls will print "I found no `\citation` commands" — this is expected and harmless on a still-blank template. It simply means there's nothing to cite yet.

7. Template Structure at a Glance

7.1 FYP-I proposal templates (Research-Based and Application-Based)

- Cover page
- Project Registration Form + Group Members table
- Summary of the Proposed Research / Project
- Plagiarism-Free Certificate, Supervisor Recommendation, Committee/Dean Approval
- Table of Contents
- Main proposal body (Introduction → References) — see Section 8 for the section list, which differs between the two tracks
- Formatting Guidelines page (for your own reference — you can delete it before final submission)

7.2 FYP-II report templates (Research-Based and Application-Based)

- Cover page, Certificate of Approval, Declaration, Dedication (optional), Acknowledgement, Abstract
- Table of Contents, List of Figures, List of Tables

- Six chapters — see Section 8 for the chapter list, which differs between the two tracks
- References (generated from references.bib)
- Appendices: Individual Contribution, Source Code, (User Manual for application-based), Plagiarism Report, Supervisor Meeting Log

8. Section-by-Section Comparison

The two tracks share the same skeleton (Introduction, related-work equivalent, methodology/design, results/testing, conclusion) but differ in what each section actually asks for:

8.1 FYP-I: Research-Based vs. Application-Based

Research-Based	Application-Based
Related Work / Literature Review	Existing Systems / Market Study
Research Questions	— (not included)
Aims and Objectives, Scope	Aims and Objectives, Scope
Proposed Research Methodology (research design, dataset, evaluation metrics)	Feasibility Study (technical / operational / schedule) + Requirements Specification (FR/NFR) + Proposed System Design (architecture, use cases)
References — inline the bibliography	References — separate references.bib

8.2 FYP-II: Research-Based vs. Application-Based

Research-Based	Application-Based
Ch.2 Literature Review	Ch.2 Existing Systems and Market Study
Ch.3 Proposed Methodology and System Design (research design, dataset, architecture, evaluation plan)	Ch.3 Requirements Specification and System Design (FR/NFR, architecture, use-case diagram, ER diagram + data dictionary, DFD/sequence diagrams, wireframes)
Ch.4 Implementation	Ch.4 Implementation
Ch.5 Results and Evaluation (metrics, discussion, threats to validity)	Ch.5 Testing and Results (test cases traced to FR/NFR, UAT, limitations)
Ch.6 Conclusion and Future Work	Ch.6 Conclusion and Future Work
Appendices: Contribution, Source Code, Plagiarism, Meeting Log	Appendices: Contribution, Source Code, User Manual, Plagiarism, Meeting Log

9. Formatting Rules Already Applied

You don't need to configure any of the following — it's already built into the .cls file:

Element	Style
Font	Times New Roman throughout
Heading 1 / Chapter titles	16pt bold (FYP-I) / centred chapter headings (FYP-II)
Heading 2	13–14pt bold
Heading 3	12pt bold italic

Element	Style
Body text	12pt, single (FYP-I) or 1.5 (FYP-II) line spacing
Margins	1 inch on all sides (FYP-I); 1.5 inch left margin for binding (FYP-II)
Figure captions	Below the figure
Table captions	Above the table
References	IEEE numbered style (ieeetr), generated automatically
Brand colour	Dark green, #153D34 — sampled from the UOL Sargodha Campus logo

10. Troubleshooting

Problem	Fix
"Undefined citations" warning after compiling	Normal on the first pass. Run the full 4-step sequence in Section 6 in order — it resolves itself.
"I found no \citation commands" from bibtex	Harmless — you haven't added any \citep{} calls yet.
A figure appears on the wrong page / far from its text	Normal LaTeX behaviour with the [H] float placement already used in these templates — figures should stay close to where you placed them. If not, try compiling twice more.
Table of contents / page numbers look wrong	Run pdflatex a second (or third) time — LaTeX needs multiple passes to settle all cross-references.
Missing package error on a local install	Update your TeX distribution (tlmgr update --self --all for TeX Live) or switch to Overleaf, which has every package pre-installed.
I accidentally deleted a \label{} or \ref{}	Compile again after fixing it — undefined-reference warnings point you to the exact missing label.

11. Pre-Submission Checklist

- No [placeholder] text remains anywhere in the document.
- All grey instruction boxes have been deleted.
- Every figure and table is referenced in the text (no "orphan" figures).
- Every entry in references.bib is actually cited at least once.
- Plagiarism/similarity check has been run and the percentage is filled in on the Declaration / Certificate page.
- All group members' names, registration numbers, and signatures are complete.
- Supervisor has reviewed and signed the relevant approval page.
- Document compiles cleanly from a fresh copy (no leftover .aux/.log confusion) — delete auxiliary files and recompile once, start to finish, before final submission.
- PDF opens correctly and the Table of Contents page numbers match the actual pages.

Questions about any of the four templates, or want a change applied across all of them (e.g. a new appendix, a different citation style, or a departmental logo update)? Just ask.